

Advanced Population Modeling : Age Structured Models

**(Restrepo & Legault 1998, Thorson et al. 2019, Fournier
and Archibald 1982) Acknowledgements: Steve Cadrin**

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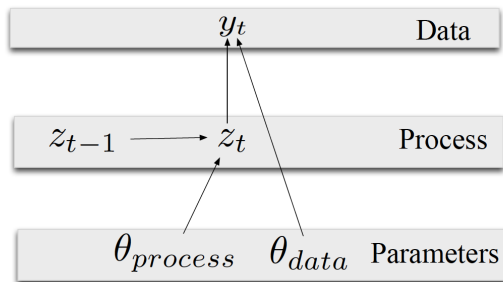
Outline

- ▶ Homework questions
- ▶ SIMULATE in TMB
- ▶ Age-structured modeling, (ASPM, SCAA)
- ▶ lab exercise: ASM fit to floundah

Process-based model

A broadly applicable approach to modeling dynamic processes in ecology

$$[\mathbf{z}, \theta_{process}, \theta_{data} | \mathbf{y}] \propto \prod_{t=2}^T [y_t | \theta_{data}, z_t] [z_t | \theta_{process}, z_{t-1}] [\theta_{process}, \theta_{data}, z_1]$$



(from Hobbs & Hooten 2015)

Populations have age structure!

Leslie Matrix of population age structure:

Vector N_t is the numbers at age. i.e.

$$N_t = \begin{bmatrix} n_{0,t} \\ n_{1,t} \\ \dots \\ n_{J,t} \end{bmatrix}$$

—

Population projection

$$N_{t+1} = AN_t = \begin{bmatrix} n_{1,t} \cdot f_1 + n_{2,t} \cdot f_2 + \dots + n_{J,t} \cdot f_J \\ n_{0,t} \cdot S_0 \\ n_{1,t} \cdot S_1 \\ \dots \\ n_{J-1,t} \cdot S_{J-1} + n_{J,t} \cdot S_J \end{bmatrix}$$

The basic age-structured fisheries model

$$N_{1,t+1} = R_{t+1}$$

$$N_{a+1,t+1} = N_{a,t}e^{-(F_{a,t}+M)}$$

$$N_{A,t+1} = N_{A-1,t}e^{-(F_{A-1,t}+M)} + N_{A,t}e^{-(F_{A,t}+M)}$$

Matrix form Age-structured Fisheries Model

$$\mathbf{N}_{t+1} = \begin{bmatrix} R_{t+1} \\ 0 \\ 0 \\ 0 \\ \dots \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & \dots & 0 & 0 \\ e^{-(F_{0,t}+M)} & 0 & 0 & \dots & 0 & 0 \\ 0 & e^{-(F_{1,t}+M)} & 0 & \dots & 0 & 0 \\ 0 & 0 & e^{-(F_{2,t}+M)} & \dots & 0 & 0 \\ 0 & 0 & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & e^{-(F_{A-1,t}+M)} & e^{-(F_{A,t}+M)} \end{bmatrix} \begin{bmatrix} N_{0,t} \\ N_{1,t} \\ N_{2,t} \\ N_{3,t} \\ \dots \\ N_{A,t} \end{bmatrix}$$

Production (and process equations)

- Change in Biomass over time
 - function of recruitment, natural mortality, growth & yield:

$$B_{t+1} = R_t \bar{w}_R + \sum_{a=k+1} (N_t e^{-M} \bar{w}_a) - Y_t$$

- (*no immigration or emigration*)

$$B_{t+1} = \sum_{a=k} (N_t e^{-(M)} \bar{w}_a) - Y_t$$

- k : age of recruitment
- What do these equations assume about the timing of the fishery relative to M ?

Production (and process equations)

- Exploitable Biomass

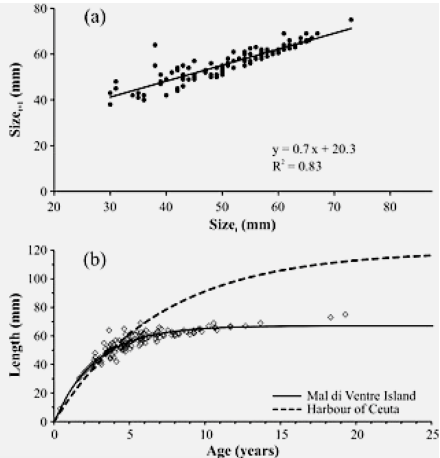
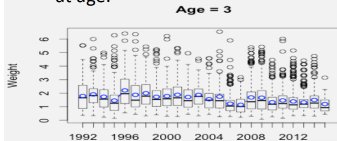
$$B_{exp,t+1} = \sum_{a=k} N_{a,t+1} v_a w_a = \sum_{a=k} (N_t v_a e^{-(M)} \bar{w}_a) - Y_t$$

- v : selectivity
- k : age of recruitment
- A simplifying assumption is to assume a pulse fishery at the end of the year:

$$F_t \cong \frac{Y_t}{B_{exp,t}}$$

Somatic Growth

- Theoretical Growth (e.g., Von Bertalanffy Model)
 - Asymptotic growth to L_{∞} or w_{∞}
 - Relationship of L_{t+1} to L_t is linear but slope is <1 , intersecting with 1 at L_{∞} .
- Empirical weight at age
 - Sample population (or fishery) to derive mean length and mean weight at age.



Survival and Mortality

- Total Mortality – negative exponential rate
 - Natural Mortality (M) + Fishing Mortality (F)
 - Natural Mortality is typically based on longevity (e.g., $M \sim 3/\text{oldest_age}$)
 - Fishing mortality at age is typically separable to fishery selectivity at age (v_a) and fishing mortality at full selectivity (F_t):

$$N_{t+1,a+1} = N_{t,a} e^{-(v_a F_t + M_a)}$$

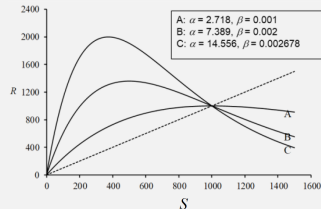
- Fishery selectivity can be approximated as the ratio of cumulative age distributions in a fishery catch (C) and a fishery-independent survey (I), which is assumed to have younger selectivity (i.e., relative selectivity):

$$v_a = \frac{\sum_{a=1}^a C_a}{\sum_{a=1}^a I_a}$$

Recruitment

- Recruitment is typically represented as
 - a density-dependent function of reproductive potential (e.g., spawning stock biomass, lagged to age of recruitment, k)
 - and a lognormally distributed deviation from the stock-recruit expectation which loosely represents environmental variability and survival from multiple early-life stages
 - e.g. Ricker model:

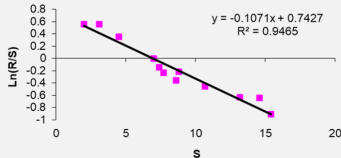
$$R_t = \alpha S_{t-k} e^{-\beta S_{t-k}} e^{\varepsilon_t}$$



Recruitment

- The stock-recruit function is the only component of productivity in a typical age-structured model that is assumed to have density-dependence (e.g., Ricker).

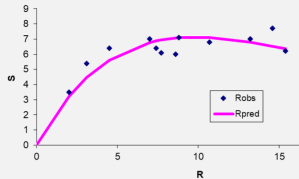
$$R_t = \alpha S_{t-k} e^{-\beta S_{t-k}} e^{\varepsilon_t}$$
$$\ln\left(\frac{R_t}{S_{t-k}}\right) = \alpha - \beta S_{t-k} + \varepsilon_t$$



Recruitment

- Age-Structured Production Models typically assume no deviation from the stock-recruit function.
- Statistical Catch-at Age models typically assume deviations from the stock-recruit function.

$$R_t = \alpha S_{t-k} e^{-\beta S_{t-k}} e^{\varepsilon_t}$$



Beverton-Holt Model & 'Steepness'

- Alternate notations:

- Original notation

$$R_t = 1 / \left(\alpha + \beta / S_{t-k} \right) = \left(S_{t-k} / \alpha S_{t-k} + \beta \right) e^{\varepsilon_t}$$

- Common notation

- $b_1 = \alpha / \beta$

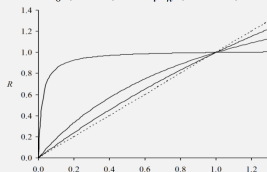
- $b_2 = \alpha$

$$R_t = \frac{b_1 S_{t-k}}{b_2 + S_{t-k}} e^{\varepsilon_t}$$

- 'Steepness'

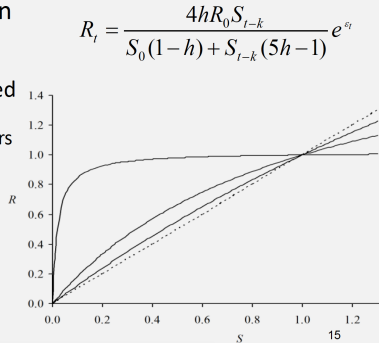
- S_0 is the maximum expected stock size (equilibrium S at $F=0$)
- R_0 is maximum expected recruitment (at S_0 , $F=0$)
- h is the steepness of the curve near the origin (proportion of R_0 predicted R at 20% S_0)
- $\alpha = 4z_{max} r_{max} / (5z_{max} - 1)$
- $\beta = [SSB_{max}(1 - z_{max})] / (5z_{max} - 1)$

$$R_t = \frac{4hR_0S_{t-k}}{S_0(1-h) + S_{t-k}(5h-1)} e^{\varepsilon_t}$$



Beverton-Holt Model & 'Steepness'

- This form of the 2-parameter Beverton-Holt model has 3 unknown parameters!
 - Steepness (h) and maximum recruitment (R_0) are typically estimated as unknown parameters.
 - Steepness (h) can be informed with priors or 'borrowed' from other stocks.
- How can we derive the expected unfished stock, S_0 , from other typically available demographic information?
 - Assumed natural mortality rate
 - Growth information

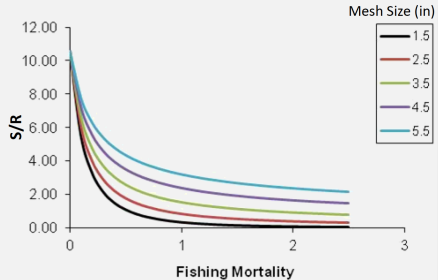


Spawner-per-Recruit Analysis

- If we have an estimate of unfished recruitment (R_0) and a calculation of Spawning-Biomass-per-Recruit at $F=0$, then unfished stock, $S_0=R_0\text{SPR}_{F=0}$:

$$S_0 = \sum_a R_0 e^{-[M(a-k)]} w_a m_a$$

- k : age of recruitment
- w : mean weight
- m : proportion mature

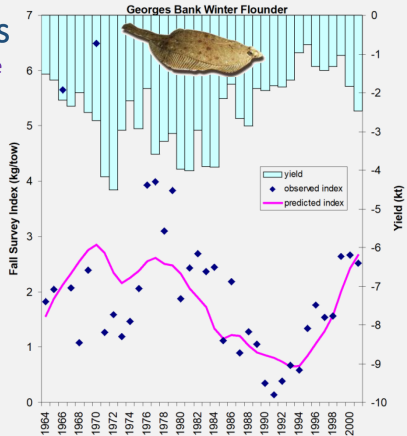


Forward Projection Models

- Pella & Tomlinson (1969) fit a **dynamic time series of biomass** to the available data by estimating Logistic growth parameters (r , K , B_{year-1}) using an observation model for a fishery or survey index of abundance:

$$I_t = qB_t e^{\varepsilon_t}$$

- Deriso (1980) developed a delay-difference model that disaggregated biomass dynamics model to the demographic components of production (recruitment, growth and survival), but specific growth and selectivity conditions apply.
- Age-structured production models do the same by estimating R_0 , h and q , assuming $SSB_{year-1} = S_0$.
 - Can either assume or estimate selectivity, M , and growth; and calculate q .

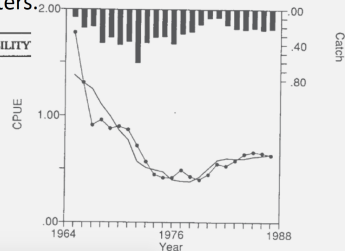


Example – Cape Hake (Hilborn 1990)

- Developed an age structured production model to relax the constraints of the delay-difference model.
- Note 'catch' is yield (kt)
- Weight and vulnerability (selectivity) can be derived from ancillary analysis or from estimated growth and selectivity parameters.



AGE	WEIGHT	VULNERABILITY
1	0.009	0.111
2	0.099	0.131
3	0.334	0.398
4	0.695	0.435
5	1.192	0.967
6	1.790	1.034
7	2.459	1.000
8	3.171	1.000
9	3.903	1.000



Example – Cape Hake (Hilborn 1990)

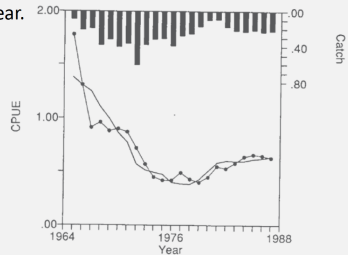


- Starting Year ($t=1$) calculations
 - Estimate recruitment in the first year ($N_{age=1,t=1}$) as an unknown parameter.
 - Assumed $M=0.3$
 - Assume an equilibrium age structure in the first year.

$$B_{exp,t=1} = \frac{I_{t=1}}{\hat{q}}$$

$$F_t \cong \frac{Y_{t=1}}{B_{exp,t=1}}$$

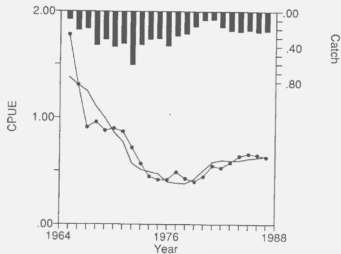
$$N_{a+1,t=1} = N_{a,t=1} e^{-(M+F_{t=1}v_a)}$$



Example – Cape Hake (Hilborn 1990)



- Forward projection of abundance at age
 - Assumed a Beverton-Holt stock-recruit relationship and maturity (m)=proportional to w for age-4+
 - Assume yield after natural mortality



$$S_t = \sum_a N_{a,t} m_a w_a$$

$$N_{age=1,t+1} = \frac{b_1 S_t}{b_2 + S_t}$$

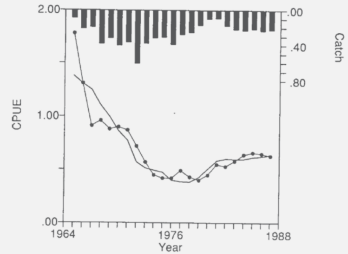
$$N_{a+1,t+1} = N_{a,t} e^{-(M+F_t v_a)}$$

Example – Cape Hake (Hilborn 1990)



- Calculation of biomass and fishing mortality each year

$$B_{exp,t} = \sum_a N_{a,t} w_a v_a$$
$$F_t \cong \frac{Y_t}{B_{exp,t}}$$

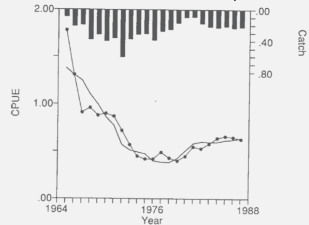


Example – Cape Hake (Hilborn 1990)



- Observation model and objective function are the same as a biomass dynamics model conditioned on yield (i.e., assumes yield is measured without error)

$$\hat{I}_t = qB_te^{\varepsilon_t}$$
$$SSQ = \sum_t [Ln(I_t) - Ln(\hat{I}_t)]^2$$



- Estimated the parameters $q=0.66$, $b_1=1.4$ and $b_2=0.28$.
- The estimated MSY is 238,000 tons (slightly less than the 257,000 tons estimated from a biomass dynamics model) – we'll cover these calculations later in the semester.

Western Atlantic Bluefin Tuna (Restrepo & Legault 1998)

- Assumed age-based parameters from ancillary analysis

Table 1. Biological parameters assumed for bluefin tuna, held constant for all years.

Age	Midyear weight (kg)	Spawning biomass (kg)	Natural mortality (yr ⁻¹)
1	4.4	0.0	0.14
2	10.6	0.0	0.14
3	20.5	0.0	0.14
4	37.5	0.0	0.14
5	56.6	0.0	0.14
6	80.6	0.0	0.14
7	111.3	0.0	0.14
8	143.0	131.3	0.14
9	176.1	164.8	0.14
10+	292.8	267.2	0.14



Western Atlantic Bluefin Tuna (Restrepo & Legault 1998)

- Maximum likelihood estimators

- Observation error

$$-\ln(L_1) = \sum_i \left[\frac{n_i}{2} \sum_t \ln(\sigma_{i,t}^2) + \sum_t \frac{1}{2\sigma_{i,t}^2} (I_{i,t} - \hat{I}_{i,t})^2 \right]$$

- Annual deviations from stock-recruit expectation:

$$N_{1,t+1} = R_{t+1} e^{\varepsilon_{t+1}} = \frac{\alpha S_t}{\beta + S_t} e^{\varepsilon_{t+1}}$$

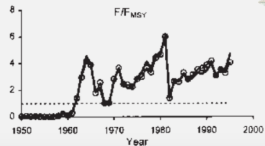
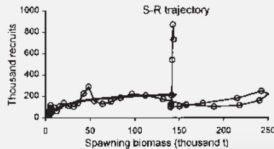
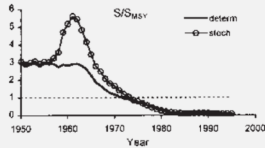
$$-\ln(L_2) = \frac{n_t}{2} \ln(\sigma_\eta^2) - \frac{1}{2} \ln(1 - \rho^2) + \frac{1}{2\sigma_\eta^2} \left[(1 - \rho^2) \varepsilon_1^2 + \sum_{t=2}^{n_t} (\varepsilon_t - \rho \varepsilon_{t-1})^2 \right]$$

- ... also included a penalty for deviation from equilibrium age structure in year-1.



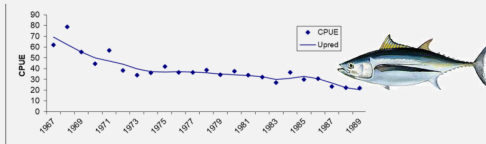
Western Atlantic Bluefin Tuna (Restrepo & Legault 1998)

- Compared two recruitment models:
 - Deterministic: expected value from stock-recruit relationship
 - Stochastic: annual deviations from stock-recruit expectations



Informative Catch Histories (still required)

- “You cannot understand how a stock will respond to exploitation unless the stock has been exploited” (Hilborn & Walters 1992).
 - ‘one-way trip’ – unreliable results
 - ‘up & down the isocline’ – more informative, but correlated estimates
 - ‘good contrast’ – several episodes of depletion and rebuilding throughout the dynamic range (0 to S_0)



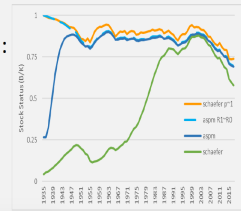
Summary

- Production
 - Age-disaggregated production models represent demographic components of production.
 - Growth is either modeled as a function of age or from empirical sampling
 - Mortality requires information on natural mortality (longevity) and selectivity
 - Recruitment is modeled as a function of stock size, typically without annual deviations (estimation of annual deviations requires age composition data)
- Time Series Estimation
 - Observation error of relative indices is assumed to be lognormal.
 - Models be simplified by calculating catchability, starting the analysis at the start of the fishery and borrowing parameter values.
 - Reliable estimates require an informative exploitation history.

Statistical Catch At Age Models

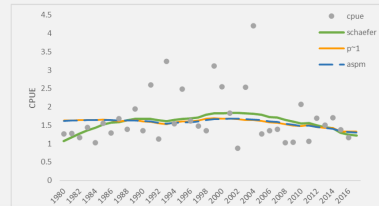
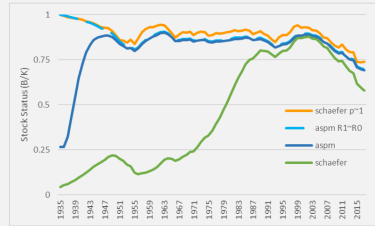
Modeling Options

- Production model (i.e., form of density dependence): logistic growth, Gompertz growth, generalized population growth, Beverton-Holt recruitment, Ricker recruitment, Shepherd recruitment, hockey-stick recruitment, ...
- Estimator (i.e., goodness-of-fit criterion): least squares, likelihood, error distributions, constraints, penalties, ...
- Starting year (data inclusion/exclusion): first year of monitored catch, first year of index series, ...
- Starting year assumption (relative to unfished conditions):
 - Estimate stock (B_1 or R_1) in the first year
 - Assume an unfished stock at equilibrium ($B_1=K$; $R_1=R_0$, $S_1=S_0$)
- Parameterization: estimate or calculate q , estimate B_1 or B_1/K (R_1 or R_1/R_0), ...



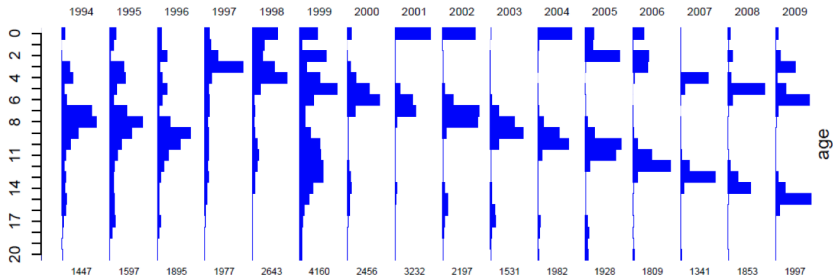
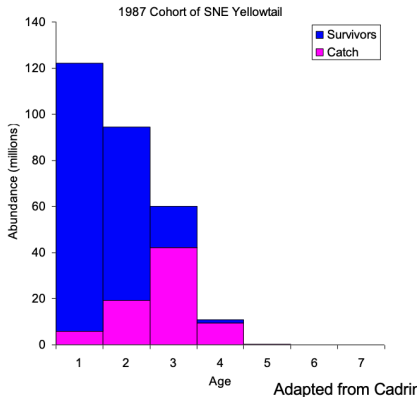
Model Components

- Population Model – process equations for recruitment, mortality, and growth
- Observation Model – equations to derive model expectations for the available data
- Statistical Model – goodness-of-fit criterion and search algorithm



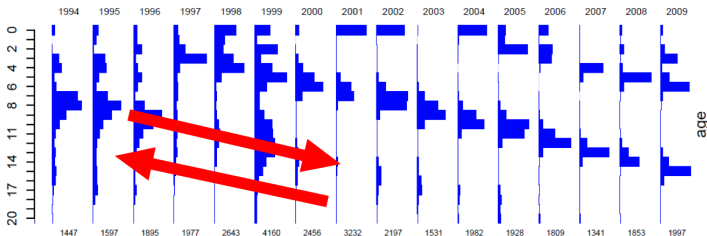
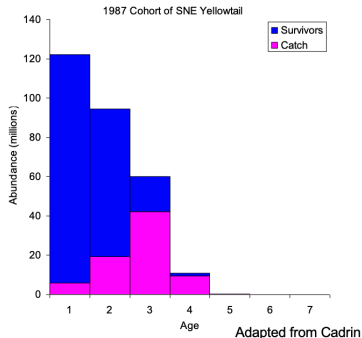
Aim is to use data on the catch-at-age to reconstruct year class size, to give a total population estimate.

Standard VPA works ‘backwards’ over the lifespan of a year-class.
whereas SCAA works ‘forwards’.



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Standard VPA works 'backwards' over the lifespan of a year-class, whereas SCAA works 'forwards'.



Klaer et al. 2011. CSIRO

Statistical Catch at Age – Process Equations

- Growth – theoretical (von Bertalanffy) or empirical (observed weight at age)
- Recruitment – usually derived from a stock-recruit with annual deviations
- Survival
 - typically continuous fishing and natural mortality (some assume ‘pulse’ fishing)
 - with estimated selectivity at age (or assumed selectivity),
 - based on abundance at age (rather than derived directly from biomass and yield).
 - For now, we’ll assume a deterministic process (no process error)

$$L_t = L_\infty [1 - e^{-k(t-t_0)}]$$

$$R_t = \frac{4hR_0S}{S_0(1-h) + S(5h-1)} e^{\varepsilon_t}$$

$$N_{t+1,a+1} = N_{t,a} e^{-(v_a F_t + M)}$$

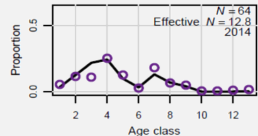
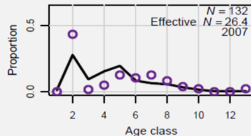
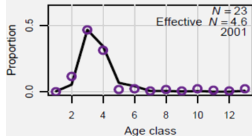
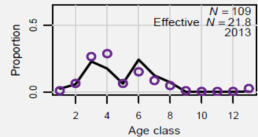
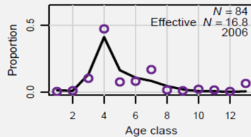
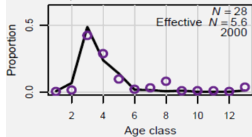
Age Composition 'Data'

- Virtual Population Analysis (and early forms of SCAA) requires complete series of catch at age (all fleets, landings and discards, all years of analysis)
 - Stratified port samples of length are raised to total landings for landings-at-length
 - Observer samples of length are raised to total discards for discards-at-length
 - Age-Length Keys are used to derive discards-at-age and catch-at-age from age-length samples.
 - Total catch-at-age is the sum of discards-at-age and landings-at-age.
 - Un-sampled strata assume data from adjacent strata



Age Composition 'Data'

- SCAA can fit directly to the data as much as possible
 - Data are not needed in every year (but samples still need to be representative)
 - Patterns can be accounted for by stratification of the model (fleets, seasons, areas, sexes, ...)



Model Expectations of Age Structure

- Doubleday (1976) predicted a time series of $\ln(C_{t,a})$ as a linear function of R_t , $F_{t,a}$, and M .
- Paloheimo (1980) predicted $\ln(C_{t,a}/E)$ as a linear function of R_t , E_t , q and M .
- **Fournier & Archibald (1982) predicted the proportion at age $P_{t,a}$ and total catch C_t from a numerical age-based population simulation of $N_{t,a}$ and F_t , v_a with assumed M .**
- Deriso et al. (1985) predicted $\ln(C_{t,a}/C_{t+1,a+1})$ from as a linear function of F and v – Catch-Age-Analysis (CAGEAN), separable VPA, Integrated Catch Analysis (ICA)

A General Theory for Analyzing Catch at Age Data

DAVID FOURNIER AND CHRIS P. ARCHIBALD

Department of Fisheries and Oceans, Resource Services Branch, Pacific Biological Station, Nanaimo, B.C. V9T 2Y8

FOURNIER, D., AND C. P. ARCHIBALD. 1982. A general theory for analyzing catch at age data.
Can. J. Fish. Aquat. Sci. 39: 1195–1207.

Model Expectations of Age Structure

- Fournier & Archibald (1982) predicted the proportion at age $P_{t,a}$ and total catch C_t from a numerical age-based population simulation of $N_{t,a}$ and F_t , v_a with assumed M .

$$C_{t,a} = N_{t,a} \frac{F_{t,a}}{F_{t,a} + M} [1 - e^{-(F_t + M)}]$$

$$C_t = \sum_a C_{t,a}$$

$$P_{t,a} = \frac{C_{t,a}}{C_t}$$

- Easily extended to predict an index of abundance and its proportion at age

Statistical Model

- Some SCAA models predict $C_{t,a}$ assuming lognormal error (with least squares of likelihood)
- Most SCAA apply Fournier & Archibald's (1982) approach of predicting C_t with lognormal likelihood and $P_{t,a}$ with multinomial likelihood

$$-Ln(L) = nLn(\sigma) + \sum \frac{[Ln(C_t) - Ln(\widehat{C}_t)]^2}{2\sigma^2}$$

$$-Ln(L) = ESS \sum_{t,a} P_{t,a} Ln(\widehat{P}_{t,a})$$

- *ESS*: effective sample size of age composition data
- ... in addition to other $-Ln(L)$ components (e.g., predicted index)

SCAA - Statistical Model

- ▶ Although the multinomial is common, other types of likelihood components are used to model composition data
- ▶ e.g. Dirichlet multinomial, logistic normal

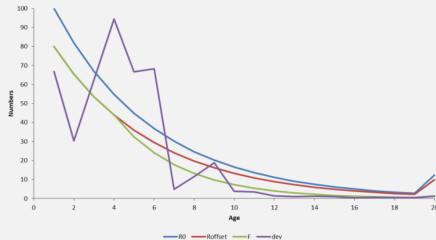
(more on this later when we discuss data-weighting)

Modeling Options

- Starting conditions:
 - Starting year - first year of monitored catch, first year of index series, first year of age composition...
 - Starting year conditions – unfished? Equilibrium?
- Fleets: aggregate, gear groups (Métiers), discards?
- Selectivity:
 - functional form or estimated for each age
 - time periods with different selectivity?
- Recruitment: related to stock and how?
- Data Weighting: purely statistical or ‘expert judgement’?
- M?
- ...

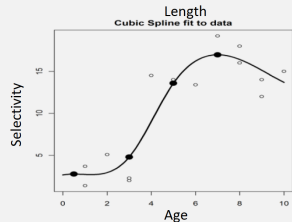
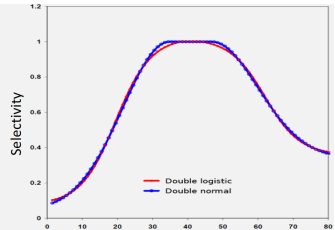
Starting Year and Conditions

- Starting year (data inclusion/exclusion): first year of monitored catch, first year of index series, first year of age composition...
- Starting year conditions:
 - Assume an **unfished stock** ($B_1=K$; $R_1=R_0$, $S_1=S_0$) or a **fished stock**
 - Assume **equilibrium** or **non-equilibrium** age structure (with non-equilibrium penalty?)



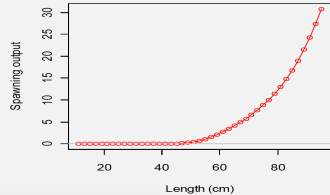
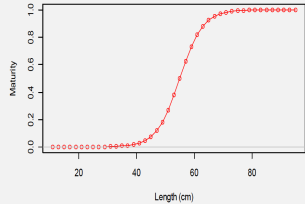
Selectivity

- Separability ($F_{t,a}=F_{full,t}v_a$) assumed for entire catch series or by fleet within periods.
- Estimate selectivity for each age v_a
- Functional Forms:
 - Full selectivity for all ages/lengths
 - Logistic – flat-topped selectivity
 - Double logistic, double normal or splines for dome-shaped selectivity
 - Random walk across ages
 - ‘Borrowed’ from a similar fleet
 - ...



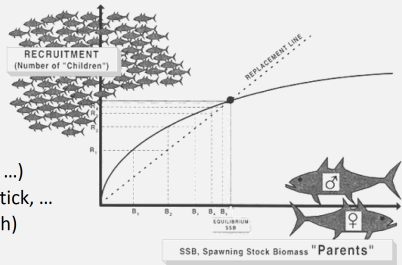
Spawning Potential

- Spawning biomass
- Female spawning biomass
- Egg production
 - fecundity = $W(a+bW)$
 - fecundity = aL^b
 - fecundity = aW^b



Recruitment

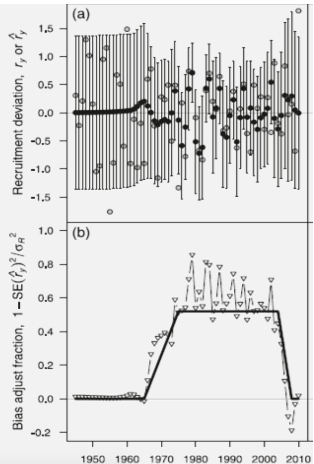
- Stock Recruit relationship
 - Measure of spawning potential (SSB, eggs, ...)
 - Beverton-Holt, Ricker, Shepherd , hockey-stick, ...
 - Deterministic or stochastic (or a mix of both)
 - Bias adjustment
- No relationship to spawners (random deviation from mean)
- Recruitment penalty in objective function?
- Autocorrelation?



Recruitment Deviations

- A penalty can be added to the objective function for deviation (r_t) from the stock-recruit expectation of recruitment
- If so, the mean recruitment prediction is biased and needs bias-correction (Methot & Taylor 2011)

$$R'_t = \frac{4hR_0S}{S_0(1-h) + S(5h-1)} e^{r_t - (\sigma_R^2/2)}$$



Recruitment

We can estimate the annual recruitments via:

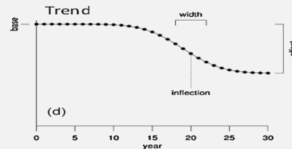
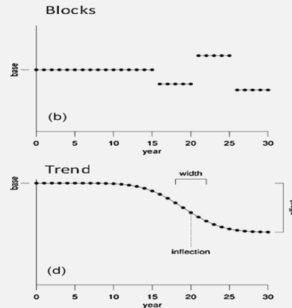
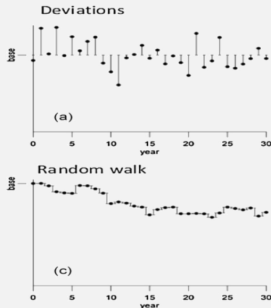
- ▶ penalized likelihood
- ▶ marginal likelihood (treat recruitments as process error and integrate them out)

(good explainer in Thorson et al. 2019)

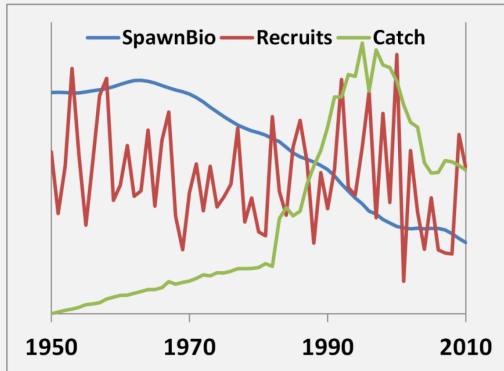
(Note that Bayesian estimation assuming distribution of annual recruitments & estimation of their variance is equivalent to marginal likelihood when the post-model predata prior for the variance is uninformative.)

Time-Varying Parameters

- Many estimated parameters (selectivity, recruitment, natural mortality, ...) can be assumed to be constant throughout the assessment time series or time-varying (Methot & Wetzel 2012)

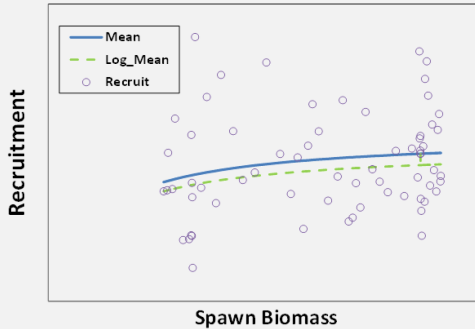


Simulation (Methot 2013)



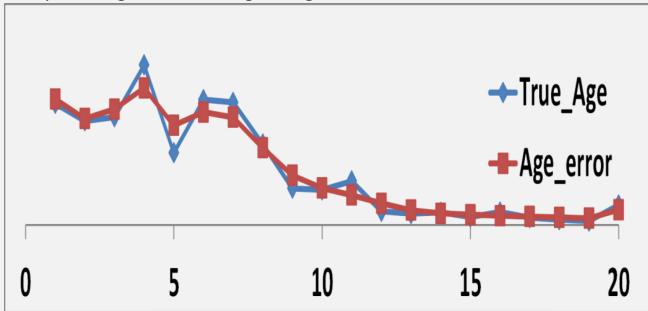
Simulation (Methot 2013)

- $h=0.7$
- $\sigma_R=0.4$
- $M=0.2$



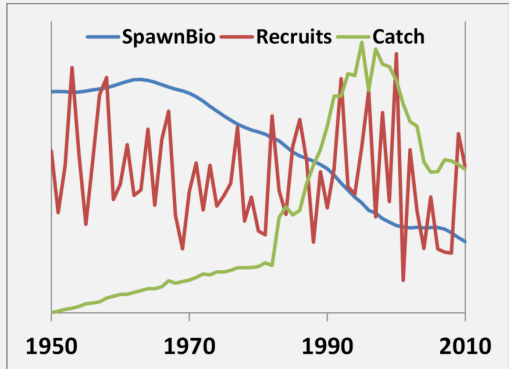
Simulation (Methot 2013)

- Imperfect ages beginning in 1971 from random observation error
- Survey of spawning biomass beginning in 1981



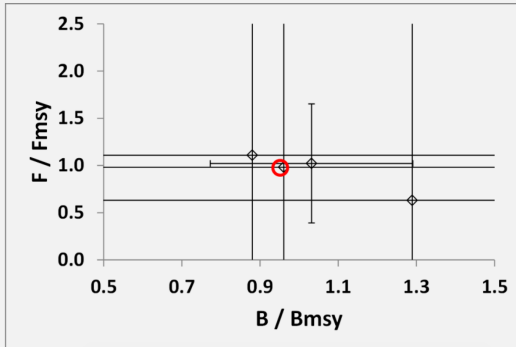
Simulation (Methot 2013)

- Attempt estimation of:
 - Steepness
 - Selectivity
 - M
 - Recruitment deviations
 - Growth
- Use informative priors in a penalized likelihood framework
- Focus on variance of model results



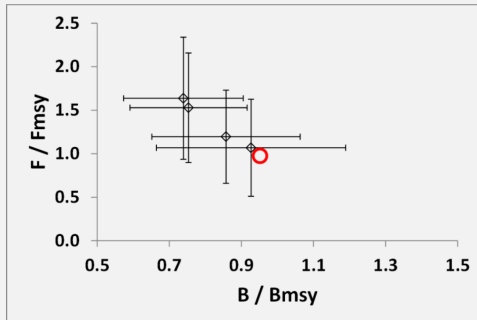
Simulation (Methot 2013)

- Results from alternative models of age-aggregate data



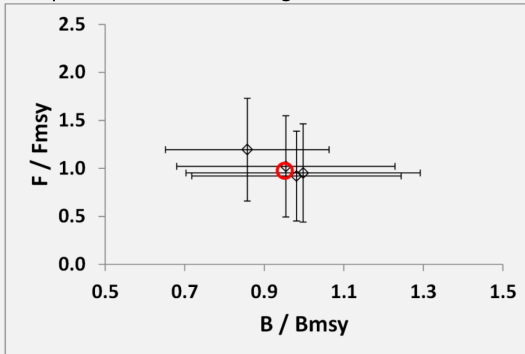
Simulation (Methot 2013)

- Results from alternative models of 'simple data' and composition data (perfect data and imprecise data)



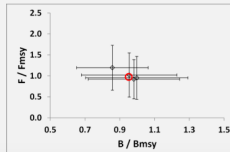
Simulation (Methot 2013)

- No process error, process error, estimating M



Simulation (Methot 2013)

- Catch time series plus some simple indicator of F is highly informative
- Three types of composition data are about equally informative
- Adding process error improved accuracy but did not greatly degrade precision
- A generalized model enables blending information from diverse data and making comparisons
- Lightly informative priors are important part of approach
- Real data must be much worse than random measurement error



Summary

- Statistical catch at age models apply the same process equations as age-structure production models, with annual recruitment deviations and fit to age composition data.
- SCAA applications range from relatively simple (single fleet, constant parameters) to extremely complex (multiple fleets, time-varying parameters)
- Fitting to age-aggregate catch and indices with lognormal error and compositional data with multinomial error is the first step in integrated modeling (later in the semester):
 - Length structure and data
 - Spatial structure and spatially-explicit data
 - Seasonal structure and seasonal data
 - Sex structure and data
 - Tagging data
- Process error can be added to process equations (next week)

Lab exercise: floundah

- ▶ Fit an age-structured production model to the flounder catch and survey data.
- ▶ Fixed input vectors of weight, maturity, and selectivity at age.
- ▶ Assume $M = 0.4$.
- ▶ Estimate h , R_0 , annual F 's & initial year recruitment (as a fraction of R_0).
- ▶ BEFORE beginning coding, map out your model, and the pseudo-code outlining the steps that will be needed to implement this model.
- ▶ *Hint* Your objective function is the same as for the biomass dynamics model (but your model predictions will differ).